Problem Statement : Development of an AI-Powered Video Interview Agent with Intelligent Proctoring 1. Background and Problem Description Recruitment teams spend considerable time conducting initial screening interviews to assess candidates' technical knowledge, communication skills, experience, and suitability for a job role. Conducting these interviews manually can be time-consuming, resource-intensive, and inconsistent across candidates. With the growing adoption of remote recruitment, another major challenge is ensuring the integrity of online interviews. Candidates may receive assistance from another person, refer to unauthorized materials, use additional devices, or obtain answers from external sources during an interview. Traditional webcam-based interviews may not provide sufficient visibility to identify all such situations. The challenge is to develop an "AI-powered Video Interview Agent" that can independently conduct interviews using the candidate's resume and the job description, interact with the candidate in real time, evaluate their responses, and monitor the interview for potential suspicious activities. Students are expected to develop a working prototype that demonstrates the complete interview lifecycle, from interview configuration and candidate onboarding to AI-led questioning, intelligent proctoring, evaluation, and report generation. 2. Objective Develop an intelligent, interactive AI agent capable of conducting structured video interviews with minimal human intervention. The system should dynamically generate relevant interview questions, communicate with candidates through voice and video, assess their answers against job-specific criteria, and provide proctoring capabilities to help recruiters identify potential interview-integrity concerns. 3. Functional Requirements A. Interview Configuration and Candidate Onboarding The system should provide an interface for recruiters to create and configure an interview. Students should develop the following capabilities: - Allow recruiters to enter the job title, job description, required skills, experience level, and other relevant job requirements. - Allow recruiters to upload a candidate's resume in PDF or DOCX format. - Extract relevant information from the resume, including skills, professional experience, education, projects, and certifications. - Allow recruiters to configure the interview duration, number of questions, question categories, and difficulty level. - If the recruiter does not specify the number of questions, the system should automatically generate a default set of 10 questions. - Provide a candidate onboarding process to check camera and microphone permissions, explain interview rules, and obtain consent for recording and proctoring. B. AI-Based Interview Question Generation The system should use an AI model or agent to generate interview questions based on the job description and candidate resume. Students should implement: - Identification of key technical skills, competencies, and responsibilities from the job description. - Matching of candidate experience and skills against the job requirements. - Generation of relevant technical, behavioral, experience-based, and scenario-based questions. - Support for recruiter-defined questions alongside AI-generated questions. - Generation of a configurable number of questions, with 10 questions as the default when no number is specified. - Appropriate variation in question difficulty according to the role and experience level. - Avoidance of repetitive, irrelevant, or duplicate questions. For example, if the job role is Software Engineer and the resume mentions Java, Spring Boot, and microservices, the agent should generate questions relevant to those skills and the candidate's claimed experience. C. Autonomous AI Video Interviewer The system should act as a virtual interviewer and conduct the interview through a video and audio interface. Students should develop the following capabilities: - Display a virtual interviewer interface that presents or speaks each question to the candidate. - Convert generated questions into speech using text-to-speech technology. - Capture the candidate's spoken answers through a microphone and convert them into text using speech-to-text technology. - Allow candidates to answer naturally through audio, with video enabled throughout the interview. - Ask the next question after the candidate finishes answering. - Generate relevant follow-up questions based on the candidate's previous responses, where appropriate. - Support configurable response time limits and interview duration. - Handle silence, unclear audio, interruptions, and temporary connectivity issues gracefully. - Maintain interview context so that subsequent questions remain relevant to the candidate's earlier answers. - Track completed, skipped, and unanswered questions. The agent should follow the configured interview structure while allowing limited adaptive questioning. If the recruiter has configured a fixed number of questions, follow-up questions should be handled according to the interview settings so that the agent does not unexpectedly extend the interview. D. Intelligent Anti-Cheating and Proctoring

One of the key challenges is to build an intelligent proctoring module that monitors the interview and identifies activities that may indicate unauthorized assistance. Students should explore computer vision, audio analysis, and event detection techniques to implement the following: - "Face presence detection: " Detect whether the candidate is visible in the primary camera feed and flag prolonged absence. - "Multiple-person detection: " Identify situations where more than one person appears in the camera frame. - "Head pose and gaze indicators: " Identify repeated or prolonged attention away from the screen as a potential concern, without treating ordinary eye movement as proof of cheating. - "Audio monitoring: " Flag possible additional voices, unusual interruptions, or other audio events that may require review. - "Suspicious activity detection: " Identify repeated patterns of potentially unusual behavior and record the relevant timestamps. - "Camera and microphone status: " Detect when the primary camera or microphone becomes unavailable or is interrupted. - "Incident logging: " Store the event type, timestamp, description, and relevant evidence for recruiter review. The system should distinguish between an observed event and an inferred suspicion. For example, looking away from the screen may have many explanations and should not automatically be classified as cheating. Proctoring alerts should support human review rather than automatically disqualifying a candidate. E. Secondary Camera Integration Using a Smartphone Students should explore an additional camera mechanism inspired by remote interviews in which candidates are required to use a second mobile phone to provide another view of their surroundings. The expected functionality includes: - Display a QR code on the interview screen that the candidate can scan using a smartphone. - Open a secure mobile web page to connect the smartphone camera to the active interview session. - Pair the secondary device with the correct interview session using a secure, temporary session token. - Stream the secondary camera feed to the interview monitoring interface. - Allow the recruiter or proctoring module to view both the primary and secondary camera feeds. - Analyze the available feeds for potential concerns, such as additional people entering the visible workspace. - Detect secondary-camera disconnection and log the event with a timestamp. - Provide clear instructions for camera placement, permissions, and connectivity. The primary camera should capture the candidate's face, while the secondary camera should provide a wider or angled view of the candidate's workspace. The system should obtain explicit consent before recording or analyzing either feed and securely handle the collected data. For the hackathon prototype, a smartphone browser-based camera connection is sufficient; students do not necessarily need to develop a separate mobile application. F. AI-Based Candidate Evaluation After the interview, the system should analyze the candidate's responses against predefined, job-specific evaluation criteria. Students should develop: - Transcription and summarization of candidate responses. - Evaluation of answers for technical correctness, relevance, completeness, and demonstrated understanding. - Comparison of answers with the required skills and competencies in the job description. - A configurable scoring rubric, such as a score from 1 to 5 for each question. - Separate assessment of technical knowledge, problem-solving, communication, and role-specific competencies where appropriate. - Identification of strengths, knowledge gaps, and areas requiring further assessment. - Evidence-based explanations for assigned scores, referring to the candidate's actual responses. - An overall assessment summary for recruiter review. The system should not rely solely on speaking speed, accent, facial expressions, or appearance to judge candidate quality. Evaluation should focus primarily on the substance of the candidate's responses and predefined job-related criteria. G. Interview Report and Recruiter Dashboard The system should provide a dashboard through which recruiters can review completed interviews and their results. The dashboard should display: - Candidate details and the associated job role. - Interview date, duration, and completion status. - Questions asked and candidate responses. - AI-generated summaries and question-wise scores. - Skills assessed and identified knowledge gaps. - Overall assessment and suggested areas for further evaluation. - Proctoring events with timestamps and descriptions. - Access to authorized interview recordings and relevant flagged segments. - Options to review, correct, or override AI-generated assessments. The report should help recruiters make informed decisions without treating AI scores or proctoring alerts as definitive hiring decisions. 4. Expected End-to-End Workflow The prototype should demonstrate the following workflow: 1. A recruiter creates an interview and uploads the job description and candidate resume. 2. The system extracts the relevant information and generates interview questions. 3. The recruiter configures the question count, or the system uses the default of 10

questions. 4. The candidate joins the interview, grants the necessary permissions, and completes the camera and microphone checks. 5. The candidate optionally connects a secondary smartphone camera by scanning a QR code. 6. The AI interviewer asks questions, listens to answers, transcribes responses, and asks relevant follow-up questions where configured. 7. The proctoring module analyzes available video and audio feeds and logs potential concerns during the interview. 8. After the interview, the AI evaluation engine scores responses against the configured rubric. 9. The system generates a structured assessment report and displays the interview results and proctoring timeline on the recruiter dashboard. 10. The recruiter reviews the evidence and makes the final hiring decision. 5. Suggested Technology Areas Students may choose suitable technologies based on their skills and the hackathon's constraints. - "AI and Large Language Models: " Interview planning, question generation, follow-up questions, response evaluation, and summarization. - "Speech Technologies: " Speech-to-text transcription and text-to-speech for the virtual interviewer. - "Computer Vision: " Face detection, person detection, head-pose estimation, and video event analysis. - "Audio Processing: " Voice activity detection and identification of potentially relevant audio events. - "Web and Real-Time Communication: " Candidate interview interface, WebRTC-based audio/video streaming, and smartphone camera integration. - "Backend and Data Storage: " Interview session management, question and response storage, assessment results, and event logs. - "Dashboard and Reporting: " Recruiter interface, evaluation reports, recording review, and proctoring timelines. Students may use existing open-source libraries and AI APIs instead of training models from scratch. The focus should be on integrating the technologies into a functional, reliable solution. 6. Expected Deliverables Each participating team should aim to deliver: 1. A working web-based AI interview application. 2. A resume and job-description upload and analysis module. 3. An AI question-generation engine with configurable question counts and a default of 10 questions. 4. A functional AI-led video interview with audio interaction. 5. A basic anti-cheating and proctoring module with timestamped incident logs. 6. A prototype for connecting a smartphone as a secondary camera. 7. An AI-based candidate evaluation engine with a configurable scoring rubric. 8. A recruiter dashboard with interview reports, scores, and proctoring alerts. 9. A demonstration of a complete interview session using a sample job description and resume. 10. Brief technical documentation describing the architecture, technology stack, limitations, and future enhancements. 7. Hackathon Evaluation Criteria Solutions may be evaluated on the following parameters: - Accuracy and relevance of generated interview questions. - Quality of AI-led interaction and follow-up questions. - Effectiveness of response transcription and evaluation. - Functionality of primary and secondary camera integration. - Ability to identify simulated proctoring events and manage false alarms. - Quality, usability, and completeness of the recruiter dashboard. - End-to-end integration, reliability, privacy, and security. - Innovation, scalability, and clarity of the final demonstration. 8. Final Expected Outcome The goal is to build a functional prototype of an autonomous AI interviewer that can conduct an interview from start to finish, adapt questions to the candidate's profile, analyze responses, and provide intelligent proctoring support. The solution should demonstrate how AI agents, speech technologies, computer vision, and real-time video communication can work together to reduce manual recruitment effort, standardize initial candidate assessment, and improve visibility into potential interview-integrity concerns. "The emphasis is on building an integrated, working solution rather than a standalone chatbot that only generates interview questions. "